

Aditi Kabra

 [aditink](https://github.com/aditink) |  [aditink.github.io](https://github.com/aditink) |  akabra@cs.cmu.edu |  +1 (607) 697 3644

My thesis is about automatically synthesizing verified control envelopes (sets of control solutions) for cyber-physical systems like trains, planes and cars. I'm excited about soundly scaling formal methods techniques using AI.

EDUCATION

Ph.D. Candidate (Computer Science), Carnegie Mellon University, USA 2020 – present

Advisors: Prof. André Platzer and Prof. Stefan Mitsch.

Bachelor of Science, Cornell University, USA 2016 – 2019

magna cum laude in Computer Science, *cum laude* in Mathematics, *cum laude* in Physics.

PUBLICATIONS

Conference Publications

- [CESAR: Control Envelope Synthesis via Angelic Refinements](#). **Aditi Kabra**, Jonathan Laurent, Stefan Mitsch, André Platzer. **TACAS 2024**.
- [Verified Train Controllers for the Federal Railroad Administration Model: Balancing Competing Brake and Track Forces](#). **Aditi Kabra**, Stefan Mitsch, André Platzer. **EMSOFT 2022. Best Paper finalist**.
- [Geometry Types for Graphics Programming](#). Dietrich Geisler, Irene Yoon, **Aditi Kabra**, Horace He, Yinnon Sanders, Adrian Sampson. **OOPSLA 2020**.

Short Papers

- [Can Large Language Models Autoformalize Kinematics?](#) **Aditi Kabra**, Jonathan Laurent, Sagar Bharadwaj, Ruben Martins, Stefan Mitsch, André Platzer. **FMCAD 2025**.

Workshop Papers

- [Online Verification of Commutativity](#). **Aditi Kabra**, Dietrich Geisler, Adrian Sampson. **TAPAS 2020**.

Preprints

- [Hybrid Game Control Envelope Synthesis](#). **Aditi Kabra**, Jonathan Laurent, Stefan Mitsch, André Platzer. arXiv preprint, 2025.

AWARDS

- Swartz Innovation Commercialization Fellow, 2023.
- Best Paper finalist, EMSOFT 2022.

SERVICE

- External/Sub-reviewer: CADE 2025, NFM 2025, TACAS 2025, FM 2024, HSCC 2024, LICS 2023, NFM 2023, FM 2022, ICCPS 2022, iFM 2022, ICCPS 2021.

- Artifact evaluation program committee: FM 2024, HSCC 2024, ASPLOS 2023, ASPLOS 2021.
- Student Ombudsperson for the CMU Computer Science Department, 2023 – present.
- CMU PLunch Organizer (principles of programming weekly lunch seminar), 2022 – 2023.

TEACHING EXPERIENCE

Teaching Assistant, Carnegie Mellon University:

- Software Foundations of Security & Privacy (Fall 2022)
- Logical Foundations of Cyber Physical Systems (Fall 2021)

Teaching Assistant, Cornell University:

- Data Mining for History and Literature (Fall 2019)
- Introduction to Analysis of Algorithms (Spring 2018, Spring 2019)
- (Consultant) Data Structures and Object Oriented Programming (Spring 2017, Fall 2017)

SYNERGISTIC ACTIVITIES

- *NSF I-Corps (Entrepreneurial Lead)*. Led an effort to develop a commercialization strategy for control envelopes through the NSF I-Corps program. We received \$50,000 in funding and interviewed a hundred industry stakeholders, learning about the market dynamics. The identified adoption barriers and opportunities inform my research priorities.
- *Patent Disclosure*. André Platzer, Stefan Mitsch, Aditi Kabra. Systems and methods for verifying train controllers. U.S. Patent Application No. US 2024/0416981 A1, filed 2 Nov. 2022; published 19 Dec. 2024.
- *Industry Experience*. Software Engineer (IC 2) at Microsoft (2020), Software Engineering Intern at Microsoft (2018, 2019), Engineering Practicum Intern at Google (2017).
- *CodeDay Labs*. In CodeDay Labs 2020 through 2024, I mentored college students to build software engineering projects over the summer in preparation for careers in technology.

REFERENCES

Prof. André Platzer

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Prof. Stefan Mitsch

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Prof. Armando Solar-Lezama

Distinguished Professor of Computing
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